



SYSTEM VULNERABILITY MANAGEMENT (CYS330)

Understanding Pert Charts

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Program Evaluation and Review Technique (PERT) chart

- Program Evaluation and Review Technique (PERT) chart is a project management tool used to plan, schedule, and coordinate tasks within a project by visually mapping activities and their dependencies
- **Benefits of a PERT Chart include;**
 1. Visualization of dependencies
 2. Critical path identification
 3. Effective time management
 4. Risk management
 5. Clear communication
 6. Resource optimization
 7. Project monitoring and control

CHALLENGES OF PERT CHARTS

LIMITATION

- The accuracy of the PERT chart depends heavily on the quality of the estimates for the activity times.

Key Features

1. **Optimistic time** (O/t_o) – Shortest duration if everything goes perfectly.
2. **Pessimistic time** (P/t_p) – Longest possible duration if challenges occur.
3. **Most likely time** (M/t_m) – Expected duration under normal conditions.
4. **Critical path** - The longest path through the network, which determines the shortest time for project completion.

SOFTWARE FOR DEVELOPING PERT CHARTS

1. LucidChart
2. SmartDraw
3. EdrawMax
4. Creately
5. Microsoft Visio

Key features

- **Activities (Nodes):** the various tasks in the project are represented by nodes. These are visual blocks representing project tasks
- **Arrows:** Indicates the dependencies between activities. They indicate which activity must take place before or after another.
- **Time Estimates per task:** Each activity is provided with estimated times for its completion.
- **Optimistic time (O):** Shortest duration if everything goes perfectly.
- **Most likely time (M):** Expected duration under normal conditions.
- **Pessimistic time (P):** Longest possible duration if challenges occur.
- **Events:** milestones or points where several activities come together or diverge
- **Critical path:** The longest path through the network, which determines the shortest time for project completion

Creating a PERT Chart

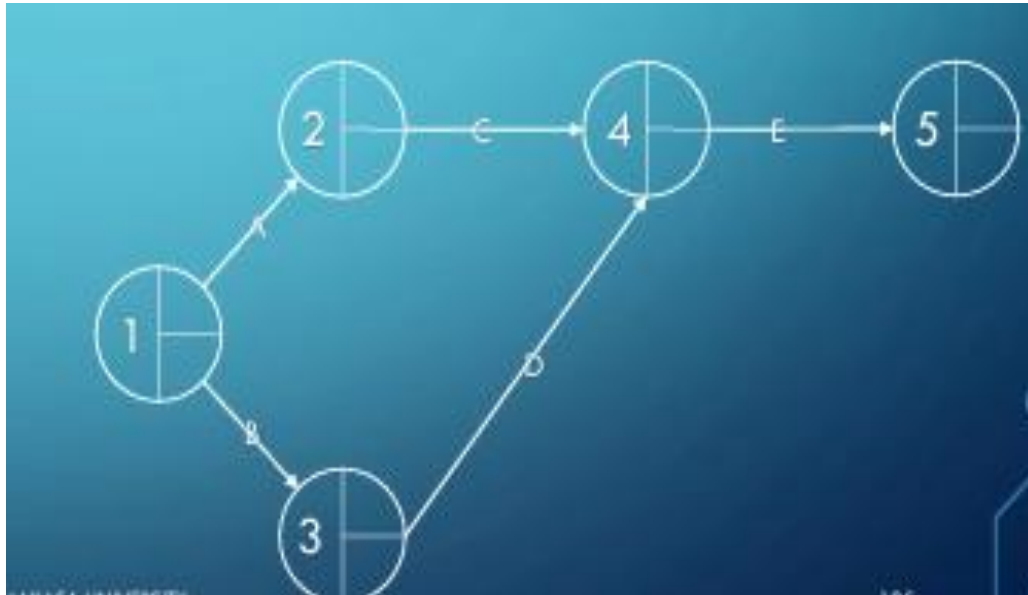
- List all Tasks: Write down every activity needed for your project
- Determine Task Dependencies: Identify which activities must precede others.
- Estimate time for each task: Assign optimistic, most likely, and pessimistic durations.
- Build the network diagram: Draw nodes for activities, connected by arrows that illustrate dependencies.
- Calculate the Critical path: Follow the longest duration path through the network to determine the minimum project completion time.
- Remember that nodes are simply circles or rectangles which display a project activity.
- Arrows connect nodes, indicating the order and dependencies of tasks.
- The network may branch with parallel activities or merge when tasks depend on multiple prior completions

FLOATS (SLACK) in a Project summary

FLOAT TYPE	MEANING	FORMULA
Total Float	Delay possible without changing project completion	$LF - EF$ or $LS - ES$
Free Float	Delay possible without delaying direct successor	$ES(\text{successor}) - EF(\text{Current})$
Independent Float	Delay possible without affecting predecessors' late finish or successors' start	Free Float Slack of tail event
Interference Float	Portion of total float interfering with successors	$\text{Total Float} - \text{Free Float}$

PERT CHART EXAMPLE 1

PERT CHART



Activities	Dependencies
A	
B	
C	A
D	B
E	C,D

PERT CHART EXAMPLE 2

- question

- 1 = True

- 0 = False

```
>>> int (True)
1
>>> int (False)
0
>>>
```

```
>>> bin(339)
'0b101010011'
>>> _
```

HOW TO CONVERT A DECIMAL NUMBER INTO BINARY SUCCESSIVE DIVISION

Now, let's use successive division to find the binary of 75.

$$75 / 2 = 37 \text{ R } 1$$

$$37 / 2 = 18 \text{ R } 1$$

$$18 / 2 = 9 \text{ R } 0$$

$$9 / 2 = 4 \text{ R } 1$$

$$4 / 2 = 2 \text{ R } 0$$

$$2 / 2 = 1 \text{ R } 0$$

$$1 / 2 = 0 \text{ R } 1$$

- We now get the binary from the bottom to the top, making it:

- 1001011

```
>>> bin(75)
'0b1001011'
>>> 0b1001011
75
>>>
```

HOW TO ADD BINARY NUMBERS

- Here are some basics of addition:

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 10$

1
+ 1
—

1 0 , this is the same as $1 + 1 = 2$
■ $2 / 2 = 1 \text{ R } 0$

```
>>> bin(1+1)
'0b10'
>>>
```

HOW TO ADD BINARY NUMBERS (CONTINUES...)

1
1
+ 1

```
>>> bin(1+1+1)
'0b11'
>>>
```

- NOTE: On the bottom we are putting the reminder (R) and carrying the number of times 2 enters into the target number

1 1 This is the same as $1+1+1 = 3$

■ $3 / 2 = 1 \text{ R } 1$

HOW TO ADD BINARY NUMBERS (CONTINUES...)

- ■ Let's take an example below:

1010 = 10

+ 1001 = 9

10011 = 19

16	8	4	2	1	Decimal
	1	0	1	0	10
	1	0	0	1	9
1	0	0	1	1	19

```
>>> bin(10)
'0b1010'
>>> bin(9)
'0b1001'
>>> bin(19)
'0b10011'
>>>
```

HOW TO SUBTRACT BINARY NUMBERS

Here are some basics of subtraction:

$$1 - 0 = 1$$

$$1 - 1 = 0$$

```
>>> bin(8-1)
'0b111'
>>> |
```

$$10 \rightarrow 02$$

$$- 1 \rightarrow 1$$

—

	1	1	
0	2	2	2
1	0	0	0
-			1
0	1	1	1

→

8	4	2	1
8			
-			1
			7

1 In base 2, will get a 2 from the nearest value

Binary Number system

understanding overflow with calculations

- A computer uses digits but there is time that a computer switches to octal, decimal and hexadecimal number system.
- Octal and hexadecimal will be skipped you can learn on your free time.
- Will look at Binary Number system
- Addition of Binary Numbers – we have two elements 0s and 1s
- Example 1

$$\begin{array}{r} 1101 \\ + 111 \\ \hline \end{array}$$

1'S & 2'S COMPLEMENT

1'S COMPLEMENT

- Obtain 1's complement of 1010.
- NOTE: In ones complement we flip the numbers, 1 becomes 0 and 0 becomes 1.
- Therefore 1's complement of 1010 is 0101

2'S COMPLEMENT

- Formula:

2's complement = 1's complement + 1

QUESTION: Obtain 1's complement of 10111010.

1s complement = 01000101

2s complement = 01000110

SIGNED MAGNITUDE

CONVERTING A BINARY

- Let's take an example, let's express (-6)

using the signed magnitude.

1. What is (+6) in binary?

8 4 2 1
0 1 1 0 = +6
↓
Sign (+)
→ Magnitude

NEGATIVE BINARY REPRESENTATION

2. The Sign (+) will change from 0 to 1
Sign (-).

8 4 2 1
Sign (-) ← 1 1 1 0

NOTE: 1 represent negative and 0 positive

2'S COMPLEMENT SUBTRACTION (CONTINUES...)

2'S COMPLEMENT SUBTRACTION

4. Finally, do the 2's complement subtraction of 39 and 25.

```
  0 1 0 0 1 1 1
+ 1 1 0 0 1 1 1
(1) 0 0 0 1 1 1 0
```

NOTE: We have a carry 1 at the end, but we are just ignoring it, meaning an Overflow as occurred.

VERIFICATION

- Is 0001110 14?

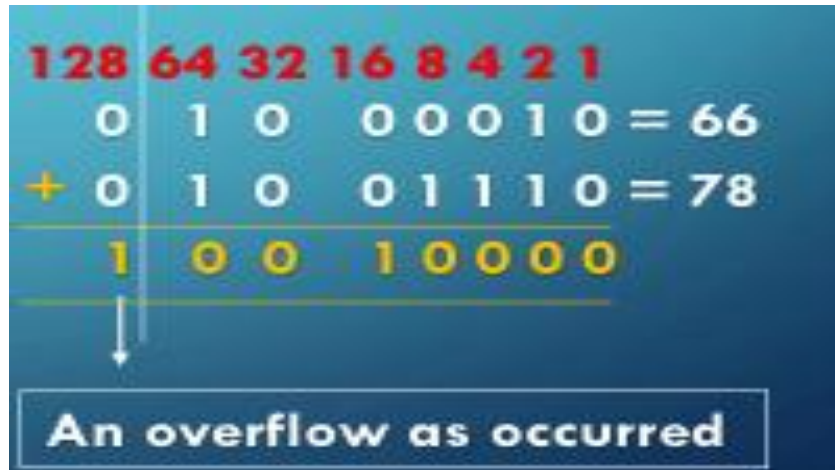
128 64 32 16 8 4 2 1

0 0 0 0 1 1 1 0

- $8 + 4 + 2 = 14$, NOTE: In 2's complement subtraction we add two binaries, think of it as $39 - 25$ is the same as $39 + (-25)$

2'S COMPLEMENT SUBTRACTION (Example 2)

1. $66 - (-78) = 144$
2. This is the same as $66 + (78)$



Intentionally left blank

2'S COMPLEMENT Addition (example 1)

1. -47 and 19

- 47 + 19 or 19 - 47 = -28

19 + (-47)

0	0	0	1	0	0	1	1	= 19
+	1	1	0	1	0	0	0	= 47's 2's comp.
1	1	1	0	0	1	0	0	

\$

128	64	32	16	8	4	2	1	
0	0	0	1	0	0	1	1	= 19
0	0	1	0	1	1	1	1	= 47
1	1	0	1	0	0	0	0	= 47's 1's compliment
+						1		
1	1	0	1	0	0	0	1	= 47's 2's compliment

2'S COMPLEMENT Addition (example 2)

1. 25 and -56

\$

$$25 + (-56) = -31$$

0	0	0	1	1	0	0	1	= 25
+	1	1	0	0	0	1	1	= 56's 1's comp.
1	1	1	0	0	0	0	0	= -31

128	64	32	16	8	4	2	1	
0	0	0	1	1	0	0	1	= 25
0	0	1	1	1	0	0	0	= 56
1	1	0	0	0	1	1	1	= 56's 1's comp.

QUESTIONS ?

